Clustering and classification for paediatric mTBI

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1 Introduction

The introduction should include background information, rationale, current status of literature, hypotheses (?) and aims/objectives.

The logical flow of my research project:

1. Paediatric mTBI can lead to serious impairments
2. Heterogeneity is a barrier to mTBI care
3. Previous attempts at characterizing this heterogeneity were limited
4. We have lots of factors and a new way to subtype over all of them!

- New data sources in depth
- SNF in depth
- Previous subtyping attempts in depth
- We are poised to address the following gaps in knowledge and objectives:
  - are there long-term subtypes of mTBI?
  - how do they differ from the regular population
  - how do the subtypes change with time?
  - how effective are the clusters for prediction (predicting WHO gets mTBI, predicting what outcomes of mTBI you will have - ANOVA across individual measures OR clustered outcomes)
  - can we validate this in other datasets?
  - what are these subtypes? classic characterization / identification methods

Concern: Our children did not all experience their mTBIs at the same time point. However, they are approximately equal in age. It is not practical to for a dataset to have a tight control on both time since injury and age, particularly outside of the context of school-level athletics. Our study assumes & tests the notion that there is heterogeneity at a persistent scale a

Mild traumatic brain injuries (mTBI) are highly prevalent in children, can result in serious and long-lasting impairments, and currently have no effective treatments. The heterogeneity of patient and injury factors is considered to be a major barrier to improving the clinical management of mTBIs. The first part of my proposed PhD work will be to apply similarity network fusion

1.1 Mild traumatic brain injury

Each **bookdown** chapter is an .Rmd file, and each .Rmd file can contain one (and only one) chapter. A chapter *must* start with a first-level heading: `# A good chapter`, and can contain one (and only one) first-level heading.

Use 2nd-level and higher headings within chapters like: `## A short section` or `### An even shorter section`.

The `index.Rmd` file is required, and is also your first book chapter. It will be the homepage when you render the book.

You can use anything that Pandoc's Markdown supports; for example, a math equation $a^2 + b^2 = c^2$.

It is also handy to account for good references, like what was done in Wang et al., 2014[1].

1.2 The Adolescent Brain Cognitive Development (ABCD) study

The Adolescent Brain Cognitive Development (ABCD) study is a long term study of pediatric development tracking patients from 21 research sites across the United States.

1.2.1 Identifying mTBI patients in the ABCD study

The ABCD study

1.3 Cluster analysis

1.3.1 Clustering techniques

1.3.2 Cluster analysis evaluation

1.3.2.1 Cluster tendency

Will not examine the Hopkins statistic for cluster tendency - "The relatively high value for H in this single-cluster case is due to a comparison of the structure of this data set so uniformly distributed random numbers, Le., data intended to be completely unstructured. With unstructured data as a comparison, virtually any structure in the test data would lead to rejection of the null hypothesis whether the data are clustered or not"[**lawson1990?**]. However even such a variation of the Hopkins statistic would not be so interesting, and would likely be expected - demographic data is random noise.

Whether or not the clusters are randomly distributed. This is challenging in the context of biomedical data - we do not expect the clusters to be randomly distributed by default (the demographic composition of any given country is far from random). In the context of mTBI, there are many facets of heterogeneity that would be interesting to examine:

- Heterogeneity of children with mTBI vs. those without
    - **Identifying the risk-of-sustaining-mtbi groups**: of all the children without a history of mTBI at baseline, is there a difference between the clustering of those that will develop an mTBI later on in the study and those who will not?
    - **Identifying the types of mTBI patients**: how does the heterogeneity of those who have mTBIs differ from the heterogeneity of those who don't have an mTBI? (Contrasted with the previous bullet point, this will help us learn more about the clusters that are formed BECAUSE of the mTBI and not because of being at a higher risk of mTBI)
        - **Identifying the types of mTBI outcomes:**
            - purely examining the clusters of patients using PPCS as the only input variable
            - can contrast mTBI PPCS vs. non-mTBI PPCS (though in that case, it's not quite a PPCS)

1.4 Similarity network fusion

Advances in technology have facilitated the collection of massive and diverse datasets. To harness the full potential of these datasets for pattern detection, prediction, and decision-making, techniques to effectively integrate diverse types of data are necessary.

Similarity network fusion (SNF), developed by Wang et al. in 2014[1], is an example of such a technique which works by first constructing similarity pairwise similarity matrices of all observations for each data type and then merging those matrices into a single network in a way that captures the full spectrum of the underlying data.

1.4.1 Advantages of SNF over other data-integration techniques

The most basic form of data-integration is to simply normalize and concatenate all available data types prior to analysis. This approach is known to reduce the signal-to-noise ratio of each individual data type[1], which is a serious concern in the context of biological data that is already very noisy to begin with. The authors of SNF also note that attempting to enhance signal by only including a pre-selected subset of data measures (features) can result in biased analyses.

Analysis of each data type prior to combination can be difficult when the results of the independent analyses don't agree with each other[1]. Circumventing this issue with consensus clustering (clustering based on the agreement of multiple different clustering algorithms) can result in the loss of valuable complementary information (i.e., the information that is not shared across data types), which is somewhat antithetical to the purpose of integrating diverse data-types in the first place.

1.4.2 Limitations of SNF

1.4.3 The SNF pipeline

1.4.3.1 The basic pipeline

The core procedure associated with SNF analyses involves the following steps:

1. Construct a similarity matrix for each pair of samples for each available data type
2. Fuse the similarity matrices using an iterative, nonlinear method (SNF)

User-defined parameters of the pipeline include:

- The data types to include
- The similarity measure to use for each data type
- Hyperparameters related to the fusion step (though the original SNF paper demonstrates strong robustness

The topic of feature selection (determining which variables to include) is not given much attention in the original description of the SNF pipeline. Though feature selection is known to be important in traditional cluster analysis due to the sensitivity of the algorithms to high-dimensionality,

1.4.3.2 The original glioblastoma multiforme case study

The authors of SNF work through a case study of applying the tool to the aggressive brain tumor, glioblastoma multiforme (GBM)[1].

1.4.4 The mathematical basis of SNF

1.5 Previous literature

1.6 Previous studies subtyping SNF

[2] applied SNF to integrate a variety of neuroimaging measures from children with neurodevelopmental disorders including autism spectrum disorder (ASD), obsessive-compulsive disorder (OCD), and attention-deficit/hyperactivity disorder (ADHD). Intriguingly, the derived clusters revealed that there were several children who were more similar to those of a different neurodevelopmental disorder than to those of their own.

The pipeline in this study consisted of

For aa right triangle, if c denotes the *length* of the hypotenuse and a and b denote the lengths of the **other** two sides, we have

$$a^2 + b^2 = c^2$$

Here is an equation.

$$f\left(k\right) = \binom{n}{k} p^k \left(1-p\right)^{n-k} (\#eq:binom) \tag{1.1}$$

You may refer to using `\@ref(eq:binom)`, like see Equation @ref(eq:binom).

2 Home

This is a book created from markdown and executable code.

See**knuth84?** for additional discussion of literate programming.

3 Summary

In summary, this book has no content whatsoever.

4 Proposed methods

Here is where I will say what I plan on doing.

5 Expected outcomes and possible pitfalls

Here is what I will talk about the things I believe may happen.

6 Identifying subjects

The TBI information of the ABCD dataset is split across a few files:

- abcd_otbi01.txt: the raw, modified Ohio State TBI screen given at baseline
- abcd_tbi01.txt: summary scores of abcd_otbi01.txt
- abcd_lpohstbi01.txt: all longitudinal follow-ups of the baseline screen
- abcd_lsstbi01.txt: summary scores of abcd_lpohstbi01.txt

All information required for identifying children at a chronic stage following mTBI as of baseline data collection is available in the abcd_otbi01.txt file.

6.1 Assigning meaningful mTBI variables

The provided variable names are non-descriptive. A renamed version of the data is used for all subsequent analyses. The code below was used for renaming the abcd_otbi01.txt file.

Script location:

```
[1] "wheeler-lab/abcd/prashanth_proj/rename-datarename_abcd_otbi01.Rmd"
```

Renaming abcd_otbi01 columns: ::: {.cell}

```r
library(tidyverse)

abcd_otbi01 <- read_delim("lab-data/abcd/4.0/trauma/abcd_otbi01.txt")[-1, ]

abcd_otbi01_renamed <- abcd_otbi01 %>%
  rename(# ever hospitalized/ER for head/neck injury?
         hosp_er_inj = tbi_1,

         # if LOC, how long?
         hosp_er_loc = tbi_1b,

         # were they dazed or have memory gap?
         hosp_er_mem_daze = tbi_1c,
```

```
    # how old were they?
    hosp_er_age = tbi_1d,

    # ever injured in a vehicle accident?
    vehicle_inj = tbi_2,

    # if LOC, how long?
    vehicle_loc = tbi_2b,

    # were they dazed or have memory gap?
    vehicle_mem_daze = tbi_2c,

    # how old were they?
    vehicle_age = tbi_2d,

    # ever injured head/neck from fall or hit?
    fall_hit_inj = tbi_3,

    # if LOC, how long?
    fall_hit_loc = tbi_3b,

    # were they dazed or have memory gap?
    fall_hit_mem_daze = tbi_3c,

    # how old were they?
    fall_hit_age = tbi_3d,

    # ever injure head/neck from violence?
    violent_hit_inj = tbi_4,

    # if LOC, how long?
    violent_hit_loc = tbi_4b,

    # were they dazed or have memory gap?
    violent_hit_mem_daze = tbi_4c,

    # how old were they?
    violent_hit_age = tbi_4d,

    # ever injure head or neck from blast?
    blast_inj = tbi_5,
```

```
    # if LOC, how long?
    blast_loc = tbi_5b,

    # were they dazed or have memory gap?
    blast_mem_daze = tbi_5c,

    # how old were they?
    blast_age = tbi_5d,

    # any other injuries with LOC?
    other_loc_inj = tbi_6o,

    # how many more?
    other_loc_num = tbi_6p,

    # how long was longest LOC?
    other_loc_max_loc_mins = tbi_6q,

    # how many were >= 30 min?
    other_loc_num_over_30 = tbi_6r,

    # what was their youngest age?
    other_loc_min_age = tbi_6s,

    # did they experience a period of multiple head injuries?
    multi_inj = tbi_7a,

    # if LOC, how long?
    multi_loc = tbi_7c1,

    # were they dazed or have memory gap?
    multi_mem_daze = tbl_7c2,

    # at what age did the effects begin?
    multi_effect_start_age = tbi_7e,

    # at what age did the effects end?
    multi_effect_end_age = tbi_7f,

    # was there another period of multiple head injuries?
    other_multi_inj = tbi_7g,
```

```
          # what was the typical effect of the injury?
          other_multi_effect_type = tbi_7i,

          # at what age did these effects start?
          other_multi_effect_start_age = tbi_7k,

          # at what age did these effects end?
          other_multi_effect_end_age = tbi_7l,

          # was there another period of multiple head injuries?
          other_other_multi_inj = tbi_8g,

          # was there another period of multiple head injuries?
          other_other_multi_effect_type = tbi_8i,

          # at what age did these effects start?
          other_other_multi_effect_start_age = tbi_8k,

          # at what age did these effects end?
          other_other_multi_effect_end_age = tbi_8l)

write_csv(abcd_otbi01_renamed, "lab-data/abcd/processed/prashanth_proj/abcd_otbi01_renamed
```

:::

Note that all variables with the same prefix (or number originally) are referring to the same injury or set of injuries.

6.2 Identifying subjects at a chronic stage post-mTBI

To identify subjects at a chronic stage post-mTBI, two intermediate pieces of information must be calculated from the original dataset:

- identifying which head injuries were mTBIs
- identifying the time since the most recent mTBI

The structure of the Ohio State TBI screen as used for ABCD data collection was not completely compatible with the most commonly used definition of mTBI[3]:

At least one of the following:

- Any period of loss of consciousness

- Any loss of memory
- Any alteration in mental state
- Focal neurological deficits

but where the severity does not exceed:

- LOC over 30 minutes
- A Glasgow Coma Scale score below 13
- Post-traumatic amnesia (PTA) greater than 24 hours

A few aspects of the data collection process in the ABCD study precluded the usage of this definition:

- No information was collected regarding the GCS
- Feeling dazed and PTA were combined into a single variable, making it unclear which combination of symptoms was present
- No information was collected regarding the duration of feeling dazed or experiencing PTA

Consequently, I used the following definition to identify mTBIs:

Any child who experienced at least one injury where that injury resulted in either:

1. Less than 30 minutes of LOC (including 0 minutes) **AND** any duration of memory loss / feeling dazed
2. LOC that is greater than 0 minutes but less than 30 minutes.

The fixed nature of the questionnaire also meant that some information about multiple mTBIs sustained in the same way or about a very large number of mTBIs throughout a child's life could have been lost (see the previous section on identifying children with a history of mTBI). For my analyses, I assume these edge cases are not common enough to meaningfully change any obtained conclusions.

The code below generates a new file which adds the following columns to the TBI screen data:

1. if a subject has experienced an mTBI prior to baseline
2. which of the reported head injuries were mTBIs
3. the time since a subject's most recent mTBI

Script location:

```
[1] "wheeler-lab/abcd/prashanth_proj/process-data/generate_abcd_otbi01_detailed.Rmd"
```

Generate detailed form of abcd_otbi01: ::: {.cell}

```r
# 2022-04-09 PV

# Summary: export list of patients who have had at least one mTBI as of baseline
# mTBI is defined as (LOC < 30 min & mem/dazed) or (0 min < LOC < 30 min)


# Load data and libraries and assign working directory
library(tidyverse)
library(magrittr)

# Import baseline data
abcd_otbi01 <- read_delim(paste0("lab-data/abcd/processed/prashanth_proj/",
                                  "abcd_otbi01_renamed.csv"),
                           guess_max = Inf)
# guess_max = Inf required to import without parsing errors (see ?readr)

# Assign injury column values to numeric
abcd_otbi01_detailed <- abcd_otbi01 %>%
  mutate(across(hosp_er_inj:other_other_multi_effect_end_age, as.numeric))


# 1. Identify if a subject has had mTBI prior to baseline (col: mTBI)
## It is not possible to properly classify mTBI in the ABCD dataset
## Here, I define an mTBI as a head/neck injury which was either:
## 1. LOC < 30 mins (LOC < 2) and memory loss (mem_daze == 1)
## 2. LOC between 0 and 30 mins (LOC == 1)

## In this section, I create an mTBI column with the values:
## 0 - definitely had 0 mTBI
## 1 - definitely had at LEAST one mTBI
## 0.5 - impossible to know if patient had at least one mTBI

abcd_otbi01_detailed %<>%
  mutate(mtbi = case_when(
    (hosp_er_loc < 2 & hosp_er_mem_daze == 1) | hosp_er_loc == 1 ~ 1,
    (vehicle_loc < 2 & vehicle_mem_daze == 1) | vehicle_loc == 1 ~ 1,
    (fall_hit_loc < 2 & fall_hit_mem_daze == 1) | fall_hit_loc == 1 ~ 1,
    (violent_hit_loc < 2 & violent_hit_mem_daze == 1) | violent_hit_loc == 1 ~ 1,
    (blast_loc < 2 & blast_mem_daze == 1) | blast_loc == 1 ~ 1,
    (other_loc_num - other_loc_num_over_30) > 0 ~ 1,
    (multi_loc < 2 & multi_mem_daze == 1) | multi_loc == 1 ~ 1,
```

```
    other_multi_inj == 1 ~ 0.5,
    other_other_multi_inj == 1 ~ 0.5,
    TRUE ~ 0))


# 2. Identify if a subject has had a moderate or severe TBI prior to baseline
## (col: moderate_or_severe_tbi)
## based on the structure of the questionnaire, this can only be done by looking at LOC
abcd_otbi01_detailed %<>%
  mutate(moderate_or_severe_tbi = case_when(
      (hosp_er_loc > 1 |
       vehicle_loc > 1 |
       fall_hit_loc > 1 |
       violent_hit_loc > 1 |
       blast_loc > 1 |
       other_loc_num_over_30 > 0 |
       multi_loc > 1) ~ 1,
    other_multi_inj == 1 ~ 0.5,
    other_other_multi_inj == 1 ~ 0.5,
     TRUE ~ 0))

# 3. Identify which injuries led to an mTBI (col: *_mtbi)

## 3-1) convert ages from approximate years to approximate months for
## subsequent calculations
abcd_otbi01_detailed$blast_age <-
    abcd_otbi01_detailed$blast_age * 12
abcd_otbi01_detailed$hosp_er_age <-
    abcd_otbi01_detailed$hosp_er_age * 12
abcd_otbi01_detailed$vehicle_age <-
    abcd_otbi01_detailed$vehicle_age * 12
abcd_otbi01_detailed$fall_hit_age <-
    abcd_otbi01_detailed$fall_hit_age * 12
abcd_otbi01_detailed$violent_hit_age <-
    abcd_otbi01_detailed$violent_hit_age * 12
abcd_otbi01_detailed$other_loc_min_age <-
    abcd_otbi01_detailed$other_loc_min_age * 12
abcd_otbi01_detailed$multi_effect_end_age <-
    abcd_otbi01_detailed$multi_effect_end_age * 12
```

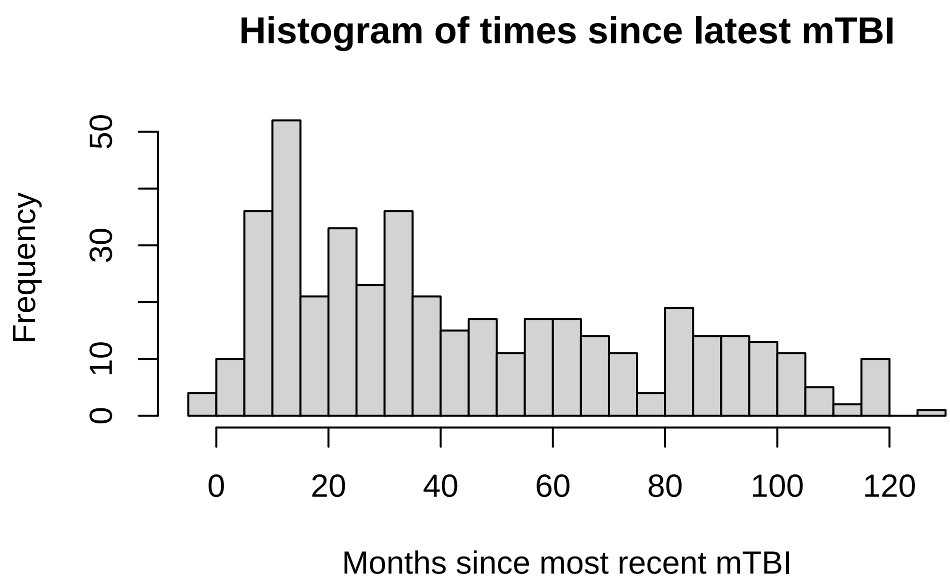
```r
## 3-2) generate columns indicating which injury types were mTBIs
abcd_otbi01_detailed %<>%
  mutate(hosp_er_mtbi = case_when(
    (hosp_er_loc < 2 & hosp_er_mem_daze == 1) | hosp_er_loc == 1 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(vehicle_mtbi = case_when(
    (vehicle_loc < 2 & vehicle_mem_daze == 1) | vehicle_loc == 1 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(fall_hit_mtbi = case_when(
    (fall_hit_loc < 2 & fall_hit_mem_daze == 1) | fall_hit_loc == 1 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(violent_hit_mtbi = case_when(
    (violent_hit_loc < 2 & violent_hit_mem_daze == 1) | violent_hit_loc == 1 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(blast_mtbi = case_when(
    (blast_loc < 2 & blast_mem_daze == 1) | blast_loc == 1 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(other_loc_mtbi = case_when(
    (other_loc_num - other_loc_num_over_30) > 0 ~ TRUE,
    TRUE ~ FALSE))

abcd_otbi01_detailed %<>%
  mutate(multi_mtbi = case_when(
    (multi_loc < 2 & multi_mem_daze == 1) | multi_loc == 1 ~ TRUE,
    TRUE ~ FALSE))


# 4. Identify time since each type of mTBI (col: *_mtbi_mpi)
## The syntax 'case_when(injury_type ~ ...)` means that variables are only calculated
## if the associated injury type actually was an mTBI
abcd_otbi01_detailed %<>%
```

```
  mutate(hosp_er_mtbi_mpi =
         case_when(hosp_er_mtbi ~ interview_age - hosp_er_age))

abcd_otbi01_detailed %<>%
  mutate(vehicle_mtbi_mpi =
         case_when(vehicle_mtbi ~ interview_age - vehicle_age))

abcd_otbi01_detailed %<>%
  mutate(fall_hit_mtbi_mpi =
         case_when(fall_hit_mtbi ~ interview_age - fall_hit_age))

abcd_otbi01_detailed %<>%
  mutate(violent_hit_mtbi_mpi =
         case_when(violent_hit_mtbi ~ interview_age - violent_hit_age))

abcd_otbi01_detailed %<>%
  mutate(blast_mtbi_mpi =
         case_when(blast_mtbi ~ interview_age - blast_age))

abcd_otbi01_detailed %<>%
  mutate(other_loc_mtbi_mpi =
         case_when(other_loc_mtbi ~ interview_age - other_loc_min_age))

abcd_otbi01_detailed %<>%
  mutate(multi_mtbi_mpi =
         case_when(multi_mtbi ~ interview_age - multi_effect_end_age))


# 5. generate column indicating time since most recent mTBI (col: latest_mtbi_mpi)
abcd_otbi01_detailed %<>%
  mutate(latest_mtbi_mpi = pmin(hosp_er_mtbi_mpi,
                                vehicle_mtbi_mpi,
                                fall_hit_mtbi_mpi,
                                violent_hit_mtbi_mpi,
                                blast_mtbi_mpi,
                                other_loc_mtbi_mpi,
                                multi_mtbi_mpi, na.rm = TRUE))
```

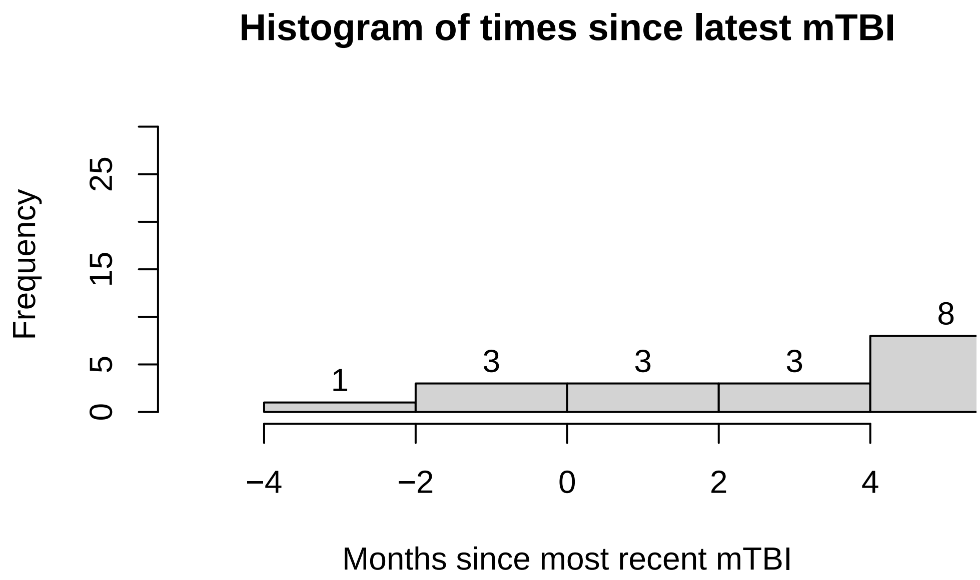
:::

The histograms below show the distribution of the time-since-last-mtbi for each subject that received an mTBI prior to baseline data collection.

```
# 6. visualize time since most recent mTBI as a histogram
## 6-1) all subjects
hist(abcd_otbi01_detailed$latest_mtbi_mpi,
     breaks=30,
     main="Histogram of times since latest mTBI",
     xlab="Months since most recent mTBI")
```




```
## 6-2) crop histogram to children that are in a non-chronic stage post-mTBI
### note that the negatives are a result of rounding used when anonymizing ages
hist(abcd_otbi01_detailed$latest_mtbi_mpi,
     breaks=50,
     xlim=c(-5,5),
     labels=TRUE,
     main="Histogram of times since latest mTBI",
     xlab="Months since most recent mTBI")
```




```r
## 6-3) count the number of children in a non-chronic stage post-mTBI
sum(abcd_otbi01_detailed$latest_mtbi_mpi < 3, na.rm=TRUE)
```

```
[1] 7
```

```r
# 7 children had an mTBI within 3 months of baseline data collection

## 6-4) evaluate mean and SD of time since latest mTBI for children at chronic stage post-

sum(abcd_otbi01_detailed$latest_mtbi_mpi >= 3, na.rm=TRUE)
```

```
[1] 424
```

```r
chronic_subject_mtbi_mpis <- abcd_otbi01_detailed %>%
    select(latest_mtbi_mpi) %>%
    filter(latest_mtbi_mpi >= 3)

max(chronic_subject_mtbi_mpis)
```

```
[1] 130
```

```r
min(chronic_subject_mtbi_mpis)
```

```
[1] 4
```

```r
sd(chronic_subject_mtbi_mpis$latest_mtbi_mpi)
```

```
[1] 32.0575
```

```r
mean(chronic_subject_mtbi_mpis$latest_mtbi_mpi)
```

```
[1] 46.06934
```

```r
# 7. Identify number of mTBIs sustained by each subject (col: mtbi_count)

## 7-1) convert NAs to 0s to include in subsequent column generation
abcd_otbi01_detailed$
    other_loc_num[is.na(abcd_otbi01_detailed$other_loc_num)] <- 0

abcd_otbi01_detailed$
    other_loc_num_over_30[is.na(abcd_otbi01_detailed$other_loc_num_over_30)] <- 0

## 7-2) generate column of mTBI counts
abcd_otbi01_detailed %<>%
    mutate(mtbi_count = (hosp_er_mtbi +
                         vehicle_mtbi +
                         fall_hit_mtbi +
                         violent_hit_mtbi +
                         blast_mtbi +
                         (other_loc_num - other_loc_num_over_30) +
                         multi_mtbi))

table(abcd_otbi01_detailed$mtbi_count)
```

```
    0      1      2      3      4      5 
11057    306    116      7      1      1 
```

Of the 431 subjects with a previous mTBI,

- 306 of them had 1 mTBI
- 116 of them had 2 mTBIs
- 7 of them had 7 mTBIs
- 1 of them had 4 mTBIs
- 1 of them had 5 mTBIs

```
write_csv(abcd_otbi01_detailed,
          "lab-data/abcd/processed/prashanth_proj/abcd_otbi01_detailed.csv")
```

A few types of injuries were handled differently from the rest in the above code:

- for `other_loc`, the questionnaire asked for a total number of injuries with LOC as well as a total number of injuries with LOC above 30 minutes.
  - If their difference was above 0, they were labeled as having had an mTBI.
- for `other_multi` and `other_other_multi`, insufficient information was provided to determine the severity of the associated injuries.

For my analyses, I consider 3 months to be the threshold at which an mTBI patient enters a chronic stage post-injury. The subject list was then defined in the following code:

Script location:

```
[1] "wheeler-lab/abcd/prashanth_proj/get-subjectsget_baseline_mtbi_subjects.Rmd"
```

```
[1] "/home/prashanth/Documents/wheeler-lab/abcd/prashanth_proj/get-subjects"
```

Export chronic-stage mTBI subject list: ::: {.cell}

```
# Load libraries
library(tidyverse)
library(magrittr)


# import data
abcd_otbi01 <-
```

```r
    read_delim("lab-data/abcd/processed/prashanth_proj/abcd_otbi01_detailed.csv",
    guess_max = Inf)


# extract patients at with mTBI
baseline_mtbi_subjects <- abcd_otbi01 %>%
  filter(mtbi == 1 & latest_mtbi_mpi >= 3) %>%
  select(subjectkey, src_subject_id)


write_csv(baseline_mtbi_subjects,
          "lab-data/abcd/processed/prashanth_proj/baseline_chronic_mtbi_subjects.csv")
```

:::

6.3 Some participant demographics

6.4 Handling twins

6.5 Identifying matched and non-matched control groups

7 Extracting input variables for subtyping

The proposed analyses for this project include the following categories of input variables:

- acute symptoms
- brain structure and function
- demographics
- psychosocial factors
- medical history

Over the next few subsections, I go collect key variables related to these categories from the NDA.

7.1 Acute symptoms

For now, acute symptoms of interest include:

- mechanism of injury
- age at which injury occurred
- presence and duration of loss of consciousness (LOC)
- presence and duration of amnesia

These variables are specifically related to the most recent mTBI. Before extracting them, I will check to see how many children experienced more than one mTBI.

7.2 Brain structure and function

Variables of interest related to brain structure and function include:

- Structural: T1-weighted multishell diffusion-weighted MRI data
- Functional: resting-state functional MRI data

The 2019 publication by Hagler et al.[4] outlines the neuroimaging data present in the ABCD dataset in detail.

7.2.1 Brain structure (!)

!Known issues

- Should I exclude automated segmentation ROIs (not a part of Anne's pilot ABCD SNF analyses)

DTI measures are provided for subcortical regions, cortical regions, and major white matter fiber tracts. The analyses in this report are only concerned with the structure of specific white matter structures, which are present in the following ROIs:

1. Fiber tracts automatically labeled by AtlasTrack [Hagler].

E.g.:

- Element: dmdtifp1_1
- Name: Average fractional anisotropy within DTI atlas tract right fornix
- Aliases: dmri_dti.full.fa_fiber.at_fx.rh, dmri_dtifullfa_fiberat_fxrh

2. Subcortical regions labeled by automated segmentation [Fischl]

E.g.:

- Element: dmdtifp1_211
- Name: Average fractional anisotropy within ASEG ROI left-cerebral-white-matter
- Aliases: dmri_dti.full.md_subcort.aseg_cerebellum.white.matter.lh, dmri_dtifullmd_subcortaseg_cereb

3. Regions of white matter sub-adjacent to cortical regions defined by anatomical parcellation [Desikan]

E.g.:

- Element: dmdtifp1_343
- Name: Average fractional anisotropy within parcellation of sub-adjacent white matter associated with cortical ROI left-medial orbito frontal
- Aliases: dmri_dti.full.fa.wm_cort.desikan_medialorbitofrontal.lh, dmri_dtifullfawm_cortdesikan_media

Detailed lists of the ROIs are present in Supplementary File 2 of[4].

- Average fractional anisotropy (FA)
- Mean diffusivity (MD)
- Average longitudinal diffusion coefficient
- Average transverse diffusion coefficient
- Fiber tract volume

For my current analyses, I include:

- non-compound regions
- FA and MD

(include rationale for why only look at these 2 and why not include GM)

The dTI data are located in the files *abcd_dmdtifp101.txt* and *abcd_dmdtifp201* (titled ABCD dMRI DTI Full Part 1 and 2 respectively). All variables follow the naming scheme 'dmdtifp1_x', where x ranges from 1 to 1185.

The dMRI variables I will include for my analyses are shown in the table below.

Variable	Segmentation method	Region type	Element name in the NDA
Fractional anisotropy	AtlasTrack	Major WM tracts	dmdtifp1_1 to dmdtifp1_35
Fractional anisotropy	Anatomical parcellation	Sub-adjacent white matter	dmdtifp_331 to dmdtifp1_401
Fractional anisotropy	Automated segmentation	WM of major brain region	dmdtifp1_211, 214, 228, 231
Mean diffusivity	AtlasTrack	Major WM tracts	dmdtifp1_43 to dmdtifp1_79
Mean diffusivity	Anatomical parcellation	Sub-adjacent white matter	dmdtifp1_402 to dmdtifp1_472
Mean diffusivity	Automated segmentation	WM of major brain region	dmdtifp1_241, 244, 258, 261

7.3 Demographics (!)

!Known issues

- I am unsure of how clustering over categorical (nominal) variables, such as race or sex, will influence outcomes
- The inclusion of these variables will simply incentivize the clustering algorithm to assign members from the same sex/race/ethnicity to the same cluster
- May be worth it to try clustering with and without these variables and check how outcomes differ

Demographic variables to be included in the analyses are:

- Age
- Sex
- Race
- Ethnicity

- Socioeconomic status
- Pubertal development status

Age and sex are commonly included in the ABCD study's data files.

7.4 Psychosocial factors

Psychosocial factors of interest include:

- family function
- prosocial behaviour
- loneliness
- healthy behaviours
- parent psychopathology

7.5 Medical history

Medical history variables of interest include:

- Number of previous head injuries
- History of headaches

7.6 Handling data types of mixed input

The original similarity network fusion paper[1] suggests using a scaled Euclidean distance for continuous variables, chi-squared distance for discrete variables, and agreement-based measure for binary variables. The paper also indicates that SNF may be used to incorporate arbitrary types of discrete (binary or categorical) data alongside continuous data, and that the compatibility of the selected data sources can be verified using normalized mutual information scores. This topic is addressed further in the methods section.

7.7 Post-concussion symptoms

8 Next steps

Here is where I will talk about what I plan on doing next.

9 Limitations

Here is where I discuss the limitations of my project.

10 Wrapping-up

Discussion

Conclusion

References

1. Wang, B. *et al.* Similarity network fusion for aggregating data types on a genomic scale. *Nature Methods* **11**, 333–337 (2014).
2. Jacobs, G. R. *et al.* Integration of brain and behavior measures for identification of data-driven groups cutting across children with ASD, ADHD, or OCD. *Neuropsychopharmacology* **46**, 643–653 (2021).
3. Blennow, K. *et al.* Traumatic brain injuries. *Nature reviews. Disease primers* **2**, 16084–16084 (2016).
4. Hagler, D. J. *et al.* Image processing and analysis methods for the Adolescent Brain Cognitive Development Study. *NeuroImage* **202**, 116091 (2019).